Fields and Waves

Modal Analysis and 3D Full-Wave Simulation of WR-90 Rectangular Waveguide

Pouria Amiri
Linkedin GitHub

October 13, 2026

Abstract

This report presents the analytical modeling and full-wave electromagnetic simulation of a standard WR-90 rectangular waveguide operating at 15 GHz. The theoretical cutoff frequencies, phase constants, and wave impedances for the first five transverse electric (TE) modes are calculated using MATLAB. Subsequently, the dominant TE_{10} mode is simulated in CST Studio Suite, and the 3D computational results are validated against the 1D analytical theory.

1 Theoretical Formulation (MATLAB)

The WR-90 rectangular waveguide has standard dimensions of $a = 22.86$ mm (broad wall) and $b = 10.16$ mm (narrow wall). For a given mode characterized by indices (m, n), the cutoff wavenumber (k_c) and cutoff frequency (f_c) are defined as:

$$k_c = \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2} \tag{1}$$

$$f_c = \frac{k_c}{2\pi\sqrt{\mu_0\epsilon_0}} \tag{2}$$

For an operating frequency ($f = 15$ GHz) higher than the cutoff frequency ($f > f_c$), the mode propagates. The phase constant (β) and the wave impedances for Transverse Electric (Z_{TE}) and Transverse Magnetic (Z_{TM}) modes are calculated via:

$$\beta = \sqrt{k^2 - k_c^2} \tag{3}$$

$$Z_{TE} = \frac{k\eta_0}{\beta}, \quad Z_{TM} = \frac{\beta\eta_0}{k} \tag{4}$$

where k is the free-space wavenumber and η_0 is the intrinsic impedance of vacuum (377 Ω). Table 1 shows the theoretical values evaluated in MATLAB.

As demonstrated, the TE_{10} mode has the lowest cutoff frequency, making it the dominant mode for the WR-90 waveguide. Modes like TE_{11} and TE_{21} are evanescent at 15 GHz since their cutoff frequencies exceed the operating frequency.

Table 1: Theoretical Waveguide Parameters at 15 GHz (MATLAB)

Mode	$\mathbf{f_c}$ (GHz)	β (rad/m)	$\mathbf{Z_{TE}}$ ($\mathbf{\Omega}$)	$\mathbf{Z_{TM}}$ ($\mathbf{\Omega}$)
TE_{01}	14.754	56.73	2087.6	67.99
TE_{10}	6.557	282.74	418.88	338.83
TE_{11}	16.145	Cutoff	–	–
TE_{20}	13.114	152.60	776.14	182.87
TE_{21}	19.740	Cutoff	–	–

Table 2: Comparison of Dominant Mode (TE_{10}) Parameters

Parameter	**MATLAB**	**CST**
Cutoff Frequency (f_c)	6.557 GHz	6.5569 GHz
Phase Constant (β)	282.74 rad/m	220.57 rad/m
Wave Impedance (Z_{TE})	418.88 Ω	443.86 Ω

2 CST Studio Suite Verification

A 3D model of the WR-90 waveguide was constructed and analyzed using CST Studio Suite. The simulation focuses on extracting the propagation characteristics of the dominant TE_{10} mode. Table 2 presents the comparison between the numerical full-wave simulation and the mathematical theory.

The extracted parameters from the 3D software closely match the 1D analytical formulas, successfully validating the physical design of the waveguide and confirming the propagation behavior of the dominant mode.